

SAINT.WTF Unity 开发者职位测试任务

要求

没有严格的截止日期，当然越早提交越好。建议完成时间为 1–2 个工作日。任务必须在 Unity 6000.3.13f1 (LTS)（必须使用此版本）中以 3D 形式完成。

下载链接：<https://unity3d.com/ru/get-unity/download/archive>

对画面效果没有要求，所有内容均可使用基础几何体（Primitive）制作，重要的是游戏机制能够正常运行。评分标准以代码架构为主。

提交格式：GitHub 仓库链接

游戏机制参考 Moon Pioneer：

<https://play.google.com/store/apps/details?id=com.norwichsidegames.tothemoon>

祝测试顺利！😊

任务内容

建筑与资源

场景中有三座建筑，分别生产三种不同类型的资源。每座建筑各有两个独立仓库：一个用于存放消耗（输入）资源，另一个用于存放生产（输出）资源。两个仓库均有容量上限。资源以三种不同颜色的长方体方块形式呈现，每种颜色对应一种资源。

每座建筑按固定时间间隔生产资源。生产链如下：

- 第一座建筑 → 生产资源 N1
- 第二座建筑 → 消耗 N1，生产 N2
- 第三座建筑 → 消耗 N1 和 N2，生产 N3

停产通知

建筑在以下两种情况下停止生产：

- 输入仓库中缺少所需资源
- 输出仓库已满

需通过 UI 文字提示玩家哪座建筑停产以及停产原因。

角色控制与背包

玩家通过虚拟摇杆控制角色（胶囊体）移动。角色拥有一个容量有限的资源背包，背包中的资源以堆叠形式显示在角色背后。

当玩家进入仓库的触发区域（**Trigger Zone**）时，可从建筑的输出仓库拾取资源，并将其送至另一建筑的输入仓库。每个资源单位的转移需要一定时间。

动画要求

所有资源移动过程均须通过线性插值动画（**Lerp**）进行可视化展示：

- 建筑 → 仓库
- 仓库 → 角色
- 角色 → 仓库
- 仓库 → 建筑

Test Assignment — Unity Developer at SAINT.WTF

Requirements

There are no strict deadlines, but sooner is better. Recommended completion time: 1–2 working days. The task must be completed in 3D, using Unity 6000.3.13f1 (LTS) (mandatory).

Download link: <https://unity3d.com/ru/get-unity/download/archive>

There are no visual requirements — everything can be built with primitives, as long as the mechanics work correctly. The assignment will be evaluated primarily on code architecture.

Submission format: a GitHub repository link

The gameplay mechanic is similar to Moon Pioneer:

<https://play.google.com/store/apps/details?id=com.norwichsidegames.tothemoon>

Good luck!

Task Description

Buildings & Resources

There are three buildings, each producing a different type of resource. Each building has two separate storage areas: one for incoming (consumed) resources, and one for outgoing (produced) resources. Both storages have a limited capacity. Resources appear as colored rectangular blocks — one color per resource type.

Each building produces its resource on a fixed time interval. The production chain works as follows:

- Building 1 produces resource N1
- Building 2 consumes N1 and produces N2
- Building 3 consumes N1 and N2, and produces N3

Production Halt Notifications

A building stops production in two situations:

- Its incoming storage lacks the required resources
- Its outgoing storage is full

The player must be notified via UI text which building has stopped and the reason why.

Character Control & Inventory

The player controls a character (a capsule) using a virtual joystick. The character has an inventory with limited capacity for carrying resources. Resources in the inventory are displayed as a stack on the character's back.

Resources can be picked up from a building's outgoing storage and delivered to another building's incoming storage when the player enters the storage's trigger zone. Each individual resource unit takes a set amount of time to transfer.

Animation Requirements

All resource movements must be visually animated using linear interpolation (Lerp):

- Building → Storage
- Storage → Character
- Character → Storage
- Storage → Building